

Name: Sakshi Padole Class: SOC-08 Batch: B

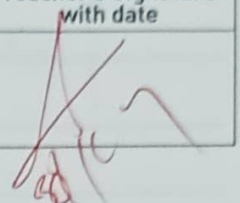
Subject: IDS Roll No.: 37 Exp. No.: 01

Name of the Experiment: _____

Performed on: _____

Submitted on: 30/01/2026

Marks
10

Teacher's Signature with date


Q.1) Define an algorithm list any four characteristics of an algorithm.

An algorithm is a finite sequence of well defined, step by step instruction used to solve a specific problem or perform a computation.

characteristic -

1. Input → An algorithm takes zero or more inputs.
2. Output → It produces at least one output.
3. Definiteness → Each step is clear, precise unambiguous.
4. Finiteness → The algorithm terminates after a finite number of steps.

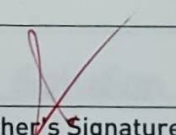
Q.2) Define data structure and abstract data type (ADT) Give one example of each.

A data structure is a way of organizing storing and managing data in memory. that it can be accessed and modified efficiently.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

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An abstract data type (ADT) is a logical description of a data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples: -

- Data structure : Array.
- ADT : stack (operation : push, pop, peek).

Q.3) Explain the process of converting a problem into a program illustrate the role of algorithms in this process.

1. problem definition - Understand requirement and constraints.
 2. Analysis: identify inputs, outputs and processing.
 3. Algorithm design: Create Step-by-step logic to solve the problem.
 4. Flowchart/ pseudocode:- Represent the algorithm visually or textually.
 5. Coding:- convert algorithm into a program using a programming language.
 6. Testing & debugging:- check correctness and remove errors.
 7. Documentations & Maintenance:- Improve and maintain the program.
- Role of algorithm -
The algorithm act as a blue print of the solution it ensures correctness, efficiency and make coding easier and error free.

Q.4) Explain linear and non linear data structure with suitable example :

a) Linear data structure:

Data elements are arranged sequentially.

- Array
- Stack
- Queue.
- linked list.

b) Non-linear data structures:

Data element are arranged in a hieratchical or interconnected manner.

Example:-

- Tree
- Graph.

Q.5) Differentiates between algorithm and program analyze their roles in SDLC.

Algorithm

program.

- | | |
|------------------------|-------------------------|
| • Step-by-step logic. | • Written code |
| • language independent | • language dependent |
| • Design level | • Implementation level. |
| • No Syntax rules. | • Follow syntax rule. |

- Role in SDLC -

- Algorithm - Used in design phase to plan the solution.
- program - Used in implement phase to execute the solution on a computer.

Q.6) Examine the relationship among data, data structure, and algorithm justify how the choice of data structure affects algorithms efficiency.

- Relationship -

- Data is raw information.
- Data structure organizes the data.
- Algorithm processes the data using data structure.

- Justification -

- The efficiency of an algorithm depends on the data structure used.

- searching in an array $\rightarrow$ Slower
- searching in a binary search tree $\rightarrow$ Faster ($O(\log n)$)

- Conclusion -

choosing the right data structure improves time and space efficiency of algorithms.

b) Explain the working of AND gate and OR gate and interpret it's truth table.

Truth table -

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Reason - Using only NAND gates, we can realize AND, OR, NOT gates.

NOR gate -

Expression - $Y = (A + B)'$

Truth table -

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Reason - Using only NOR gates, all logic Functions can be built.

Q.2

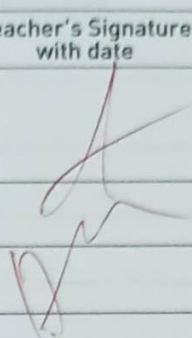
Name: Sakshi Padole Class: SOC-08 Batch: B

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Name of the Experiment: _____

Performed on: _____

Submitted on: 17-02-2026

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Q.1. Sequential organization & characteristics of linear data structures.

It is a method of storing data elements in a continuous memory location where elements in a continuous memory location are arranged one after another in a fixed order.

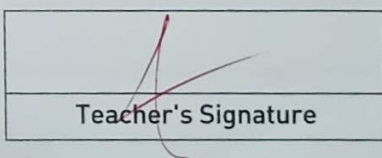
Characteristics:

- elements are stored in contiguous memory locations.
- each element has a unique index or position.
- Direct access to elements is possible using indexing.
- Data is arranged in a linear sequence.
- Size is usually fixed.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

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Q.2 Array as an abstract data type (ADT).

ADT is collection of elements of the same data type started in contiguous memory locations & accessed using an index.

- Components of ADT -

elements - Data items of the same type.

Index - Position used to access elements

Size - Total number of elements.

- Basic operations on array -

Traversal - Visiting each element.

Insertion - Adding a new element.

Deletion - Removing an element.

Searching - Finding an element.

Q.3] Insertion, deletion & searching in a 1D array.

→ Consider an array:

$A = [10, 20, 30, 40]$

Insertion: insert at position 2:

shift elements right $\rightarrow A = [10, 20, 25, 30, 40]$

Deletion - Delete elements at position 3:

shift elements left $\rightarrow A = [10, 20, 25, 40]$.

Searching - To search 25.

Compare sequentially until found.

25 is found in index 2.

Q.4] Advantages & limitations of arrays.

• Advantages:

- Fast direct access using index ($O(1)$)
- Simple & easy to implement
- Efficient memory usage for fixed size data.
- Good for storing homogeneous data.

• Limitations -

- Fixed size
- Insertion & deletion are time-consuming.
- Wastage of memory if size not useful.

• Efficiently comparison:

- Insertion - $O(n)$ due to shifting.
- Deletion - $O(n)$ due to shifting.
- Searching - $O(n)$ for linear search.
 $O(\log n)$ for binary search (sorted array)

Q.5] 2-dimensional array for student academic data:

2D array - Rows represent students
Columns represent marks/subject.

Ex:-

Student	Math	Science	English
S ₁	85	90	88
S ₂	78	82	80

- Accessing data -

- marks[0][1] → science marks.
- Transversed uses nested loops.

- Efficient manipulation -

- Direct indexing allows fast access.
- Nested loop helps in processing total & average.

- Extension to n-dimensional arrays -

- Image processing (3D arrays for RGB)
- Scientific Simulations.
- ML & data analysis.
- Game development.